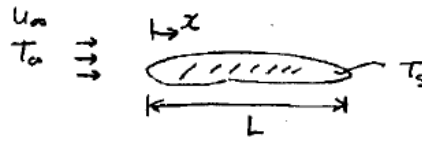


12. Boundary layer similarity

Back to the same question



Q) $\tau(x)$? $\bar{\tau}$? $h(x)$? $\bar{h}$?

A) Physical observation suggests

$$\tau = f(u_\infty, \rho, \mu, L, x) \quad h = f(u_\infty, \rho, \mu, L, k, c_p, x)$$

$$\bar{\tau} = f(u_\infty, \rho, \mu, L) \quad \bar{h} = f(u_\infty, \rho, \mu, L, k, c_p)$$

Furthermore, dimensional analysis (PI theorem) yields

- $\bar{C}_f = \frac{\bar{\tau}}{\rho u_\infty^2 / 2} = f(\text{Re}_L)$, $C_{f,x} = f(\text{Re}_x, x^*)$ $x^* \equiv \frac{x}{L}$
- $\bar{Nu}_L = \frac{\bar{h}L}{k} = f(\text{Re}_L, \text{Pr})$, $Nu_x = f(\text{Re}_x, \text{Pr}, x^*)$ ← Need to show these! (see lecture notes 10SA and 10SB)

If the functions f s contain all the parameters correctly, the analysis is ok.

Boundary layer equations confirm that there is no missing parameter

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$u(x, y=0) = v(x, 0) = 0$$

$$u(x, y=\infty) = u_\infty$$

$$T(x, 0) = T_s$$

$$T(x, \infty) = T_\infty$$

Conclusion:

- Dimensionless heat transfer coefficient (Nu_x) is a function of Re_L , Pr , and dimensionless location x/L as dimensionless shear stress ($C_{f,x}$) is a function of Re_L , and dimensionless location x/L

$$Nu_x = f(\text{Re}_x, \text{Pr}, x^*), \quad \bar{Nu}_L = f(\text{Re}_L, \text{Pr})$$

$$C_{f,x} = f(\text{Re}_x, x^*), \quad \bar{C}_f = f(\text{Re}_L)$$

- Reynolds number $Re_L = \frac{u_\infty L}{\nu} = \frac{\rho u_\infty^2 / L}{\mu u_\infty / L} = \frac{\text{inertia force}}{\text{viscous force}}$
 - Nusselt number $Nu_L \equiv \frac{hL}{k}$ $Nu_L \sim \frac{h\Delta T}{\frac{k}{L}\Delta T} \sim \frac{\text{convection HT}}{\text{conduction HT with } (\underline{v}=0)}$
 - Prandtl number $Pr \equiv \frac{\nu}{\alpha} = \frac{(\mu/\rho)}{(k/\rho c_p)} = \frac{\mu c_p}{k} \sim \frac{\text{momentum diffusivity}}{\text{thermal diffusivity}} = \frac{\text{viscous transport}}{\text{thermal transport}}$
- Later we will show that $\frac{\delta}{\delta_t} = \frac{\text{boundary layer thickness}}{\text{thermal BL thickness}} = Pr^n$

[Dimensionless BL EQ]

$$\begin{aligned}
 (A) \quad u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} &= \frac{1}{Re_L} \frac{\partial^2 u^*}{\partial y^{*2}} & u^* &\equiv \frac{u}{u_\infty} \\
 (B) \quad u^* \frac{\partial T^*}{\partial x^*} + v^* \frac{\partial T^*}{\partial y^*} &= \frac{1}{Re_L Pr} \frac{\partial^2 T^*}{\partial y^{*2}} & x^* &\equiv \frac{x}{L} \\
 & & v^* &\equiv \frac{v}{u_\infty} \\
 & & y^* &\equiv \frac{y}{L} \\
 & & T^* &\equiv \frac{T - T_s}{T_\infty - T_s}
 \end{aligned}$$

These equations show the importance of Re and Pr. Also, they confirm

$$\overline{C_f} = f(Re_L), \quad C_{f,x} = f(Re_x, x^*) \quad \text{and} \quad \overline{Nu_L} = f(Re_L, Pr), \quad Nu_x = f(Re_x, Pr, x^*)$$

because

$$\begin{aligned}
 \text{Eq.(A)} &\rightarrow u^*, v^* = f(Re_L, x^*, y^*) \\
 &\rightarrow C_{f,x} = \frac{\tau}{\frac{1}{2} \rho u_\infty^2} = \frac{2}{Re_L} \left. \frac{\partial u^*}{\partial y^*} \right|_{y^*=0} = f(Re_L, x^*) \\
 \text{Eq.(B)} &\rightarrow T^* = f(Re_L, Pr, x^*, y^*) \\
 &\rightarrow Nu_x = \frac{hx}{k} = \frac{-k \left. \frac{\partial T}{\partial y} \right|_{y=0}}{T_s - T_\infty} \frac{x}{k} = \left. \frac{\partial T^*}{\partial y^*} \right|_{y^*=0} = f(Re_L, Pr, x^*)
 \end{aligned}$$

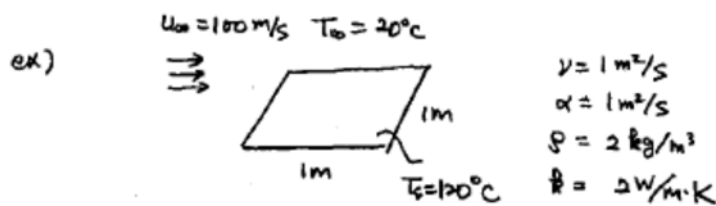
[Reynolds analogy]

Comparison of Eq.(A) with Eq.(B) shows If $Pr = 1, u^* = T^*$ (Note BCs are same)

$$\rightarrow \frac{\partial u^*}{\partial y^*} = \frac{\partial T^*}{\partial y^*} \rightarrow \frac{1}{2} Re_L C_{f,x} = Nu_x \quad \left(\text{Note } C_{f,x} = \frac{2}{Re_L} \frac{\partial u^*}{\partial y^*} \bigg|_{y^*=0} \text{ and } Nu_x = \frac{\partial T^*}{\partial y^*} \bigg|_{y^*=0} \right)$$

Reynolds analogy:

- If $Pr = 1, \frac{1}{2} C_f = \frac{Nu}{Re_L Pr} = St \quad (Pr = 1)$
- If $Pr \neq 1, \frac{C_f}{2} = St Pr^{\frac{2}{3}} = Nu Re_L^{-1} Pr^{-\frac{1}{3}} \quad (0.6 < Pr < 60) \leftarrow \text{Modified Reynolds analogy}$
(Chilton-Colburn analogy)
- Reynolds analogy relates C_f with Nu (dimensionless velocity with dimensionless temperature, momentum transfer with heat transfer)
- This analogy tells you how to measure a velocity field with a thermocouple, how to predict heat transfer coefficient from the result of velocity simulation, etc.
- Origin of the analogy: similarity of boundary layers $\delta \sim \delta_t$



Total drag force = 1N

a) q ? (total heat transfer rate)

$$q = \bar{h} A (T_s - T_{\infty}) \quad h? \rightarrow Nu?$$

$$\bar{C}_f = \frac{F_D}{A} = \frac{1 \text{ N}}{1 \text{ m}^2} = 1 \text{ N/m}^2$$

$$\bar{C}_f = \frac{\bar{C}_f}{\frac{1}{2} \rho u_{\infty}^2} = 10^{-4}$$

$$(6.103) \quad \frac{C_f}{2} = St \cdot Pr^{\frac{2}{3}} \quad (Pr = 1)$$

$$\frac{\bar{C}_f}{2} = \bar{Nu} Pr^{-\frac{1}{3}} Re_L^{-1}$$

$$Re_L = \frac{u_{\infty} L}{\nu} = 100 \quad \therefore \bar{Nu} = \frac{\bar{C}_f}{2} Re_L = 5 \times 10^{-3}$$

$$\bar{Nu} = \frac{\bar{h} L}{k} \quad \therefore \bar{h} = \bar{Nu} \frac{k}{L} = 0.01 \text{ W/m}^2\text{K}$$

$$\therefore q = 0.01 \times 1 \times 100 = 1 \text{ W}$$